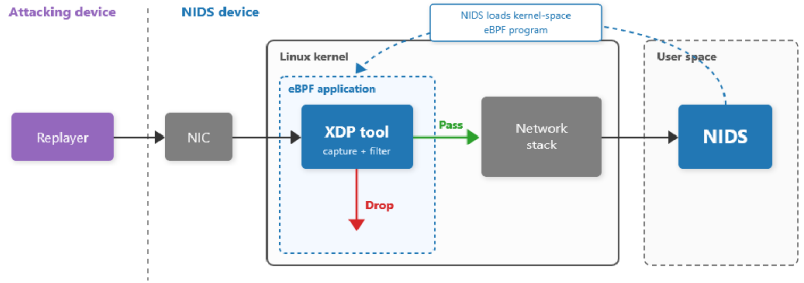
6. **In-Kernel DDoS Detection with eBPF/XDP**

Towards Multi-Layer Intrusion Detection Systems



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Background/motivation: Modern DDoS attacks can overwhelm a network before a user-space intrusion detector has time to react. In our recent work, XDP performs very fast packet filtering, but the actual detection logic runs in user space: XDP checks a blocklist, forwards packet metadata to user space, and only drops a source after the detector has classified it as malicious. *Can we move part of the detection directly into XDP and stop obvious attacks earlier, while keeping more sophisticated analysis in user space?*

You will learn about: eBPF/XDP and Linux kernel networking and system design; high-performance packet processing; BPF maps and streaming statistics; DDoS detection; experimental evaluation of detection accuracy, throughput, latency, and resource overhead.

Scope & example earlier work to build on:

The project will extend an existing Go-based NIDS combining user-space anomaly detection with eBPF/XDP enforcement. A first step could be to implement lightweight packet/byte-rate or heavy-hitter detection directly in XDP, followed by a hybrid design in which XDP handles obvious volumetric attacks while suspicious traffic is escalated to the user-space Isolation Forest or an Autoencoder model. This is particularly interesting for highly distributed attacks, where simple per-source blocking can be ineffective. Another problem to investigate is which IP to block once a window is flagged as malicious.

Other selected relevant references / sources:

•Existing Go/eBPF NIDS from the preceding MSc work, https://github.com/Bi9River/goflowmeter + preprint based on recent master thesis.
•CIC-DDoS2019 dataset for controlled traffic replay and comparison.
•Farasat et al., *SmartX Intelligent Sec: A Security Framework Based on Machine Learning and eBPF/XDP;* RW combining ML-based detection with eBPF/XDP.
•Høiland-Jørgensen et al., *The Express Data Path: Fast Programmable Packet Processing in the Operating System Kernel*, CoNEXT 2018.
•CIC-DDoS2019 dataset and taxonomy.
•Previous statistical NIDS thesis using Count-Min Sketch / HeavyKeeper as a lightweight baseline.